

Reset Neutral

Team Name: Team Bright

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Date: 5/10/2026

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Executive summary

ResetNeutral will be an all-inclusive webpage for those looking to get into fighting games. The fighting game genre is well known for having a high barrier of entry and being extremely difficult to master. This is only exacerbated by the fact that many games are locked behind having to purchase it, and devoting one's time to it to practice and figure out if one likes it. Our solution to this is to have a central repository of fighting game knowledge and character move-sets that is freely and publicly accessible on the internet. Additionally, we will have a "training ground" to allow people to practice inputting combos for characters and see whether or not they want to spend the time mastering said game or character. Our solution will provide a centralized teaching tool that will allow players of different skill levels to learn and practice as well as allowing interested people to try out a game before buying it and sinking their time into it.

Introduction

1.1 Motivation & Problem Statement:

The fighting game genre is well known for having a high barrier of entry and being extremely difficult to master. This is only exacerbated by the fact that many games are locked behind having to purchase it, and devoting one's time to it to practice and figure out if one likes it.

1.2 Objectives, Scope, and Goals Achieved:

Objectives:

1. Website framework correctly serves up webpages
2. Home screen webpage with menus are complete and navigable
3. Webpages with general/common information mostly complete
4. A selection of some characters from some of the more popular games finished and integrated
5. Training ground with simple input detection and combo/moveset recognition
6. Training ground allows selection of games and characters

Scope: What was built was a quick functional website with a selection of three games and over seventy characters. Each game and character have their own webpage and each character's page has a built in training ground with input detection with moveset and combo recognition.

Goals Achieved: What was achieved was a properly functioning website with an innovative training arena with low-latency input handling that properly mimics a fighting game all contained within the website.

1.3 Proposed Solution:

Our solution to this is to have a central repository of fighting game knowledge and

character move-sets that is freely and publicly accessible on the internet. Additionally, we will have a “training ground” of sorts to allow people to practice inputting combos for characters and see whether or not they want to spend the time mastering said game or character.

1.4 Key Highlights:

Reset Neutral boasts a unique way of testing fighting games by containing its own input validation method within the website itself. With this input validation it's possible to test input moves and combos from within the website. Additionally, Reset Neutral provides a way to save a user's progress and mastery with an account feature. Reset Neutral also provides a way to scale data by limiting the number of database fetches, and allowing the browser to handle much of the computation.

Project Specification & Management

2.1 The Project Specification (The AI's Blueprint + Link):

The project specification document is a 28 page, AI-ready document where we detailed the entirety of our product. Information included is an overview of the product, the tech stack, the project file structure, the data models, the project phases, the API reference in full, the UI/UX guidelines, the error handling standards, the non-functional requirements, and the environment variables.

Spec Sheet:

<https://github.com/CS4485-Team-Bright/CS4485-Project-ResetNeutral/blob/main/Documents/ResetNeutral%20Spec.docx>

2.2 Maintaining the "Single Source of Truth":

When major changes or significant project developments occurred, then the project specification sheet would be updated to reflect those changes. Thus allowing the AI model to adapt the project accordingly. Minor changes that did not warrant a whole project specification change would either be updated manually or run with smaller individual AI prompting and then integrated into the larger whole. These changes would either be saved up to make one major change or were added to the project specification sheet when they occurred without running the AI on the whole sheet again.

2.3 Key Milestones:

- We moved from project proposal and brainstorming to designing on February 20th which is where we designed the figma UI, API framework, and database format.
- The completion of the technical spec was on February 27th.
- The first functional MVP was on March 6th deployed and hosted on Netlify.
- The transition from Netlify to Cloudflare was on April 8th due to limitations on Netlify and the less constricting nature of Cloudflare.

2.4 Team Roles & Leader Rotation:

Team Roles

Role	Team Member	Responsibilities
Backend Infrastructure Developer	Adam M. Jackson	Database & Auth Lead (Supabase)
Frontend UI Developer	Zachary Karanja	Combo Logic & Visualization
Frontend Input Developer	Josh A. McKone	Input Game-Loop Integration
Frontend UI Developer	Alexis Vazquez	Global UI/UX Development
Frontend Input Developer	Eric M. Wilhoit	Input State Machine Lead
Backend Infrastructure Developer	Luis C. Gutierrez	Data Architect / API Lead

Team Leader Rotation

Duration	Team Leader
02/16 - 02/27	Zachary Karanja
03/02 - 03/13	Alexis Vazquez
03/16 - 03/27	Eric M. Wilhoit
03/30 - 04/10	Josh A. McKone
04/13 - 04/24	Adam M. Jackson
04/27 - 05/08	Luis C. Gutierrez

System Architecture & High-Level Design

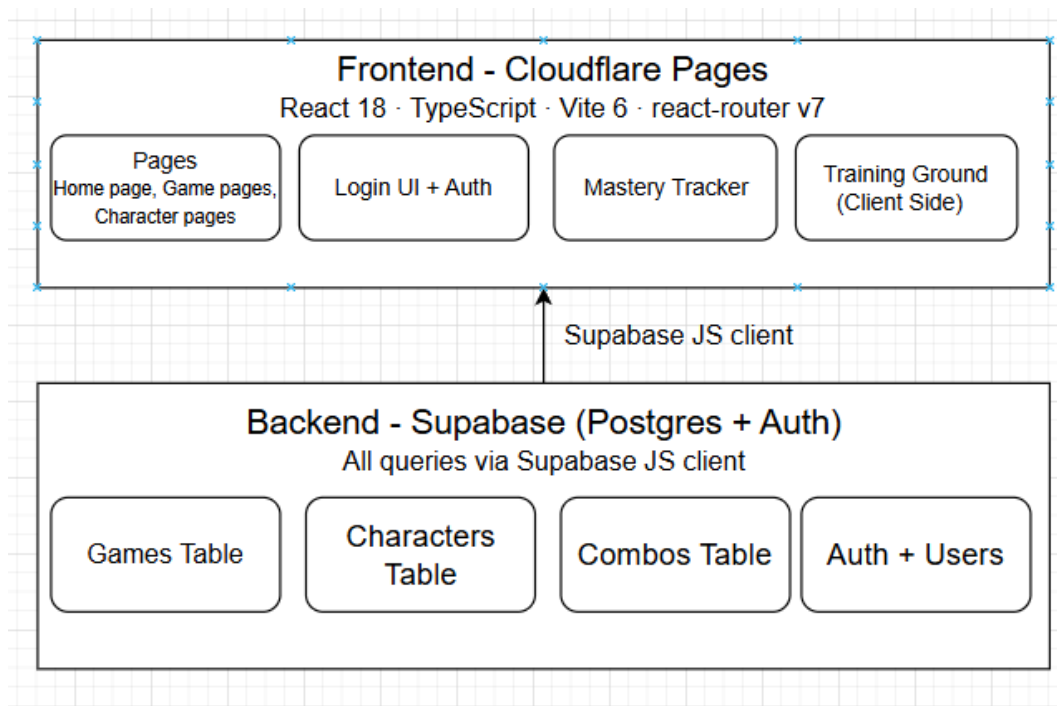
3.1 System Overview:

Our proposed solution was a website where the user interface would contain a main menu to act as a home page, specific game pages that would list fighters of a specific game, specific character pages that would contain their moveset, and a training ground that would interact with the user to input correct button combinations, additionally having methods to travel between these pages smoothly. Our backend involved a database that would contain data from the wiki to serve to browsers while the user browses the website. This would be accessed by the API to be transferred from backend to frontend. The backend would also be responsible for handling user accounts and saving their data.

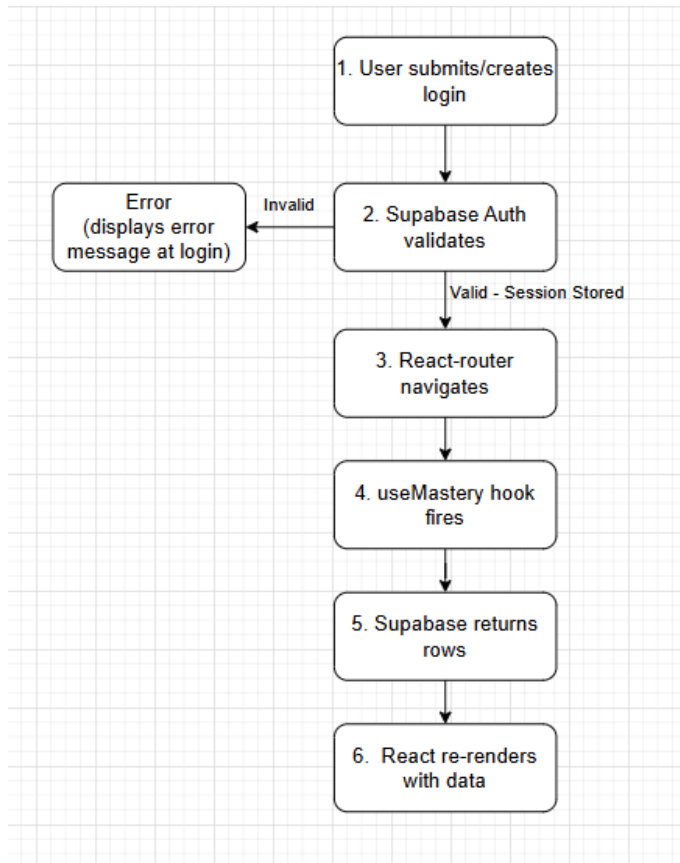
<https://github.com/CS4485-Team-Bright/CS4485-Project-ResetNeutral/blob/main/Documents/CS4485%20Project%20Proposal.pdf>

3.2 Architecture Diagrams:

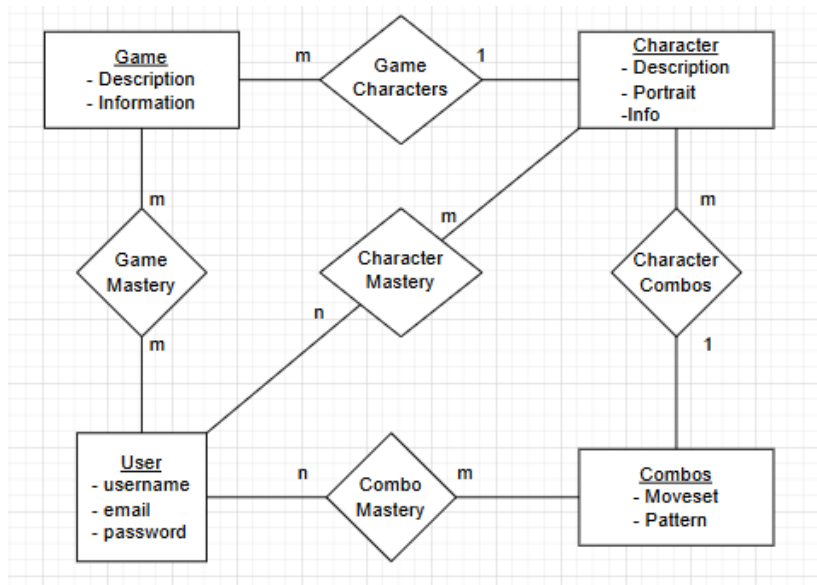
Architecture:



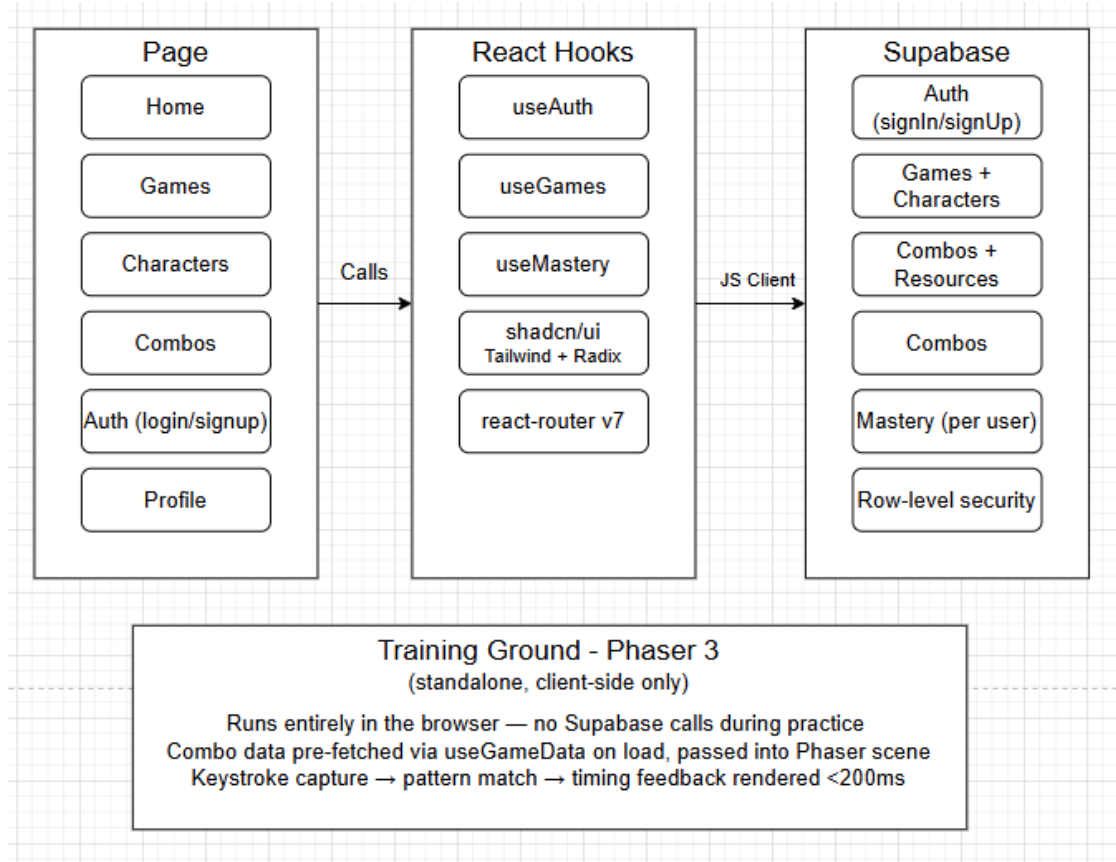
Data Flow:



ER Diagram:



Component Interactions:



3.3 Key Architectural Decisions and Trade-offs:

Our project tackles scalability by focusing on limiting the amount of database fetches and handling as much computation from within the local browser. This was done by handling inputs and combos clientside and caching combos on the users device. This was the middle ground of what we could do for the database

<https://docs.google.com/document/d/1-sLd1deg3SGvRxw8hU2mgkpiABpi6kH5929fFhKIDM/edit?usp=sharing>

Implementation & AI Orchestration

4.1 Technologies Used:

- Figma
- Cloudflare Pages - using Javascript, CSS, and HTML
- Supabase - quick prototyping and user authentication
- GitHub - Project management and version control
- Claude Code with Cursor IDE for development

4.2 AI Build Strategy:

Various members of the team used different techniques over the course of the project to develop specific parts. One of the major instances was in the development of the site's frontend. AI was used extensively to get the visuals and site layout done. This was done by prompting the AI with our specification sheet and additional prompting after that for smaller details and modifications. This prompt consisted of describing the type of site we were looking to create along with all the major functional parts that needed to be there. We then tested different AI models, all using the same prompt, to see which one could generate a design that was closest aligned with our specifications. The selected design was decided upon through a team vote, in our case it was a design generated by Claude.

4.3 Human-in-the-Loop Interventions:

While a detailed log was not kept of instances that required specific human intervention, there were some important instances where human intervention was necessary. Some of the major ones involved were in the database where the AI tended to hallucinate characters and information from the various fighting games. These hallucinations included character names and spellings, combo data, and character descriptions and move information. These all required some significant human intervention and data verification throughout that stage of the project. Some other minor instances were in the development of the Phaser library as earlier versions weren't as AI-friendly as later ones, setting up the website pipeline initially, and times where team members ran out of tokens and had to debug manually as a result.

4.4 Codebase:

Github Repository:

<https://github.com/CS4485-Team-Bright/CS4485-Project-ResetNeutral>

README:<https://github.com/CS4485-Team-Bright/CS4485-Project-ResetNeutral/blob/main/README.md>

Documents Folder containing Meeting Minutes, Weekly Reports, The Project Proposal, and full Spec sheet:

<https://github.com/CS4485-Team-Bright/CS4485-Project-ResetNeutral/tree/main/Documents>

System Performance & Validation

5.1 Testing and Validation:

We used Vite with Github Actions to test builds before they were made live to the website. This also allowed us to automate posting to the site whenever a push was made to the main branch in the repository. We also used Pytest for API testing and unit testing.

5.2 Technical Performance Metrics:

We used Supavisor for testing connections to the database with values going all the way up to 1 million. For site loading metrics we used Cloudflare Observatory that tested for things like TTFB and page load.

5.3 AI Model Performance (if included):

For this project, AI was not involved in the system itself, only to assist with the creation of it.

Lessons Learned

6.1 Insights Gained:

One of the major takeaways from the shift to AI-orchestrated engineering was just how quickly development could be done. The use of AI greatly increased productivity and reduced the overall time that needed to be spent developing. We also learned how important and useful it was to have an overall specification master sheet for the AI to develop the project alongside us.

6.2 Lessons from Challenges:

Most technical challenges encountered involved either limitations in the AI or from limitations from outside APIs and sources. For example, there were some early struggles getting data from other open source projects and repositories to fill out character and game information. We also had some issues with site limits in our earlier hosting platform. This one was fixed by switching hosting services. Most of the other problems were solved with manual work on our end. This took the form of changing how we were gathering the data from the sites by modifying our scripts, manually verifying and modifying the data in the database, and reading over documentation and doing bug fixes by hand when AI failed to do so for us.

6.3 Skills Developed:

Overall, the team's skills in AI greatly improved and several team members learned some new skills that will be useful for future projects. Everyone had developed some experience with AI over the course of the project. Some members learned about API calls, website scripting, input handling, database management and creation, and various other technical skills of that nature. Teamwork was also very good throughout the project with members of the team regularly picking up for others when the need arose with equal work load sharing. Additionally, responsiveness and planning for issues was pretty decent throughout the project's timeline.

6.4 Individual Reflection:

Luis Chantel Gutierrez:

I believe this was a good challenge with something I was unfamiliar with and worth working on in the future. Although I was mostly learning proper API protocols as I went, AI was super helpful for starting out (with proper oversight and tests, of course).

Adam Morton Jackson

I worked quite in depth with our database and in doing so learned more about databases and the use of AI in populating the database. Formatting and filling a database with AI allowed me to finish the database much quicker and begin linking it to the front-end.

Zachary Karanja

I am proud of our output for this project and find the premise of input handling an interesting challenge. I learned a lot about website development through this, creating a website UI and deploying it as well. These are topics I generally knew about but I've never really hosted something outside of local hosting. The use of AI helped me bridge the gaps in my knowledge quite seamlessly.

Joshua Alexander Mckone:

I learned a lot of interesting technical skills throughout the course of the project. One of the big ones was input handling on a website. I had experience with it before but only on locally run applications and never through a web browser. That taught me a lot about limitations and overcoming them. Some smaller skills learned was database manipulation, caching, and tool chains for site development and deployment.

Alexis Vazquez

Working on this project taught me a lot about how AI can be used to enhance my workflow by filling in gaps in my knowledge, and how it can increase the speed of learning unfamiliar tools. I have had previous experience with frontend development, however using UX tools like Figma and hosting platforms like Cloudflare were new to me. I had only worked on existing codebases, and had not had experience developing frontend from scratch. It feels so rewarding seeing what started as an idea, become a real, fully-fledged website that people can visit and use. Fighting games and its community are one of my biggest passions, so I felt very grateful for the opportunity to work on a project related to them.

Eric Mitsuji Wilhoit

I learned a massive amount about creating and hosting a live website throughout this project. I also greatly benefitted from the chance to use AI workflows on a real project that had the scope to make this effort worthwhile. I got that experience from using Cursor on my own as well as writing the first draft of the project spec. I also appreciated the opportunity having this large scale project gave to gain familiarity with testing and CI/CD. I learned how to use Vite to set up route and smoke tests for the project and how to use GitHub actions to run them automatically on deployment, which will definitely be helpful in the future if I do more web development. I'm glad we were able to complete a project everyone on the team was passionate about with a

good looking website as an end result.

Conclusion & Future Work

7.1 Summary of Key Accomplishments:

Of our original objectives, we managed to accomplish most of them. We were able to add the three games we originally proposed and most of their respective characters. This includes their names, pictures, descriptions, information, combos, and combo images. For the combos, we also successfully implemented a “training ground” system where users of the site can practice the movesets. As part of this practice, we also added a mastery system where users can sign up with an account and then track their mastery progress. The mastery system also keeps track of successful combo streaks, character moveset mastery, and overall game mastery level. The account system is also one of the portions of the project that we successfully implemented with successful backend integration and load testing.

7.2 Future Enhancements:

For future improvement, there are two major sources of enhancements we would be interested in doing. The first is to open the project up to the community as an open source endeavor. This will allow the community to run the project and make modifications beyond what we are capable of, as well as allowing the project to continue even if we are unable to continue it ourselves. The second is to continue adding more fighting games and characters. We had several other games in our stretch goals but were unable to add them due to time constraints and focused instead on getting the project itself set up and settled. There are dozens of other fighting games out there and hundreds of characters to be added. We implemented the project in such a way that adding them shouldn't be too difficult, but will just take extensive time due to the differences between the games. We have also considered adding more features to the site such as a more extensive controller support amongst other things.

References

- Supercombo and Dustloop
- Phaser Game Library and Documentation
- Claude Code with Cursor IDE
- Supabase documentation
- Cloudflare Pages documentation

- REST API documentation

Acknowledgements

We want to thank Professor Sridhar Alagar for giving us the opportunity to do this project this semester as well as guidance throughout the project itself.

We also want to thank Adarsh Gella for being extremely helpful in developing the project, giving us guidance when we were stuck, and overall for being a pleasure to work with over the course of the semester.

From our team, Thank you both very much for everything you have done for us over the semester and we wish you the best for the summer.

Also, thank you to everyone in the team for working hard over the semester and for getting the project into its current working state.